

Probability & Statistics

Moment generating function (MGF)

$\forall t \in \mathbb{R}$

$$M_X(t) = E(e^{tx})$$

provided $E(e^{tx})$ exists.

$$= \sum_{x_j \in \mathcal{R}_X} e^{tx_j} f(x_j)$$

if X discrete

$$= \int_{\mathcal{R}_X} e^{tx} f(x) dx$$

if X is continuous.

i) $M_X(0) = 1$

$$\left. \frac{d}{dt} M_X(t) \right|_{t=0} = E(X)$$

$$\begin{aligned} \left. \frac{d^j}{dt^j} M_X(t) \right|_{t=0} \\ = E(X^j) \end{aligned}$$

Quartiles / Median

X : r.v.

$F_X(x)$: CDF of X .

m = median ; $m \in \mathbb{R}$

$$F_X(m) = \frac{1}{2}$$

Transformation of r.v.s.

x	-2	-1	0	1	2	3
$p(x)$	p_{-2}	p_{-1}	p_0	p_1	p_2	p_3

consider $Y = x^2$

Discrete case : easy!

Let x be a continuous r.v. with pdf $f_x(x)$
and CDF $F_x(x)$.

$$Y = X^2 :$$

Q: What is the pdf of Y ??

Let $G(y)$ be the CDF of Y and $g(y)$ be the pdf of Y .

$$\begin{aligned} G_Y(y) &= \text{Prob}(Y \leq y) && \forall y \in \mathbb{R} \\ &= \text{Prob}(X^2 \leq y) \\ &= \text{Prob}(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}) \end{aligned}$$

$$G_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

$$\frac{d}{dy} G_Y(y) = g_Y(y) = f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + f_X(-\sqrt{y}) \frac{1}{2\sqrt{y}}$$

$$g_Y(y) = \frac{1}{2\sqrt{y}} [f_X(\sqrt{y}) + f_X(-\sqrt{y})] \quad ; \quad y \geq 0$$

Ex: X is a c.b. r.v. with pdf

$$f_X(x) = \begin{cases} \frac{2x}{R^2} & 0 \leq x \leq R \\ 0 & \text{o.w.} \end{cases}$$

$$Y = X^2$$

$$g_Y(y) = \frac{1}{2\sqrt{y}} \left[\frac{2\sqrt{y}}{R^2} + 0 \right] \quad 0 \leq y \leq R^2$$

$$g_Y(y) = \frac{1}{R^2}$$

$$= 0$$

$$0 \leq y \leq R^2$$

o.w

Ex: $X \sim U[0,1]$

$$Y = -\frac{1}{\lambda} \log(1-X)$$

for $\lambda > 0$

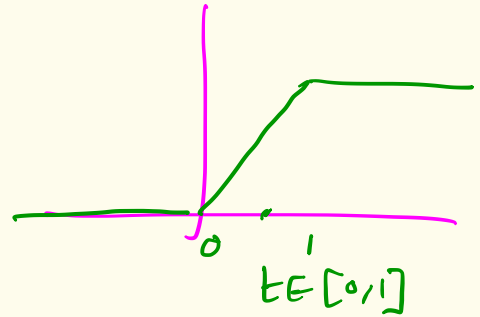
Let $G_Y(y)$ be the CDF and $g_Y(y)$ be the pdf of Y .

$$\begin{aligned}
 G_Y(y) &= \text{Prob}(Y \leq y) \\
 &= \text{Prob}\left(-\frac{1}{\lambda} \log(1-x) \leq y\right) \\
 &= \text{Prob}(\log(1-x) \geq -\lambda y) \\
 &= \text{Prob}(1-x \geq e^{-\lambda y}) \\
 &= \text{Prob}(x \leq 1 - e^{-\lambda y})
 \end{aligned}$$

$$G_Y(y) = F_X(1 - e^{-\lambda y})$$

$$G_Y(y) = 1 - e^{-\lambda y}$$

$$\begin{cases}
 g_Y(y) = \lambda e^{-\lambda y} & y > 0 \\
 = 0 & \text{o.w.}
 \end{cases}$$



$$F_X(t) = t$$

$$\boxed{\forall y > 0}$$

Ex: $X \sim U[0,1]$

$$Y = -\frac{1}{\lambda} \log(X)$$

$$\lambda > 0$$

$$G_Y(y) = \text{Prob}(Y \leq y)$$

$$= \text{Prob}\left(-\frac{1}{\lambda} \log X \leq y\right)$$

$$= \text{Prob}(\log X \geq -\lambda y)$$

$$= \text{Prob}(X \geq e^{-\lambda y})$$

$$= 1 - F_X(e^{-\lambda y})$$

$$= 1 - e^{-\lambda y}$$

$$\left[\begin{array}{l} f_Y(y) = \lambda e^{-\lambda y} \\ \quad = 0 \end{array} \right. \quad \left. \begin{array}{l} y > 0 \\ \text{o.w.} \end{array} \right]$$

